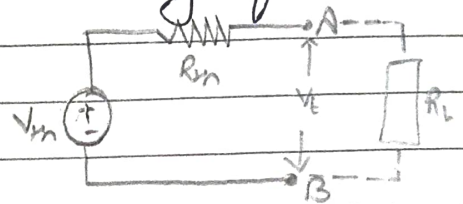


Q1. A network has an open circuit terminal voltage of 12.6V. When a  $2\Omega$  load is connected, the terminal voltage drops to 11.8V. Find the  $R_{th}$  and predict the terminal voltage for a 5A load.

Sol:- Given:  $V_{th} = V_{oc} = 12.6V$

$$V_t = 11.8V, \quad R_L = 2\Omega$$

$$I_L = \frac{V_t}{R_L} = \frac{11.8}{2} = 5.9A$$



also,  $V_t = V_{th} - I_L R_{th}$

$$R_{th} = \frac{V_{th} - V_t}{I_L}$$

$$\Rightarrow R_{th} = \frac{(12.6 - 11.8)}{5.9}$$

$$\Rightarrow R_{th} = 0.136\Omega$$

If  $I_L = 5A$  then,

$$\begin{aligned} V_t &= V_{th} - I_L R_{th} \\ &= 12.6 - (5 \times 0.136)V \\ &= 11.92V \end{aligned}$$

Q2. For  $L = 220\mu H$  and  $C = 470\mu F$ , find the frequency at which  $X_L = X_C$ .

Sol:- Since,  $X_L = X_C$

$$\text{or } \omega L = \frac{1}{\omega C}$$

$$\text{or } \omega = \frac{1}{\sqrt{LC}}$$

$$\text{or } \omega = \frac{1}{\sqrt{220 \times 10^{-6} \times 470 \times 10^{-6}}}$$

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$$\omega = \frac{1}{3.216 \times 10^{-4}}$$

$$= 3109.45 \text{ rad/s.}$$

$$\text{So, } f_0 = \frac{\omega}{2\pi}$$

$$[\omega = 2\pi f]$$

$$= \frac{3109.45}{2\pi} \text{ Hz}$$

$$= 494.88 \text{ Hz}$$

Q3. An inductor in a switching circuit sees +10V for 6 $\mu$ s and then -V<sub>2</sub> for the remaining 4 $\mu$ s of a 10 $\mu$ s period. Find V<sub>2</sub> for steady-state operation.

Since from Volt-second balance :-

$$\int_0^T v_L(t) dt = 0$$

$$\text{So, } 10(6 \times 10^{-6}) - V_2(4 \times 10^{-6}) = 0$$

$$\Rightarrow V_2 = \frac{10 \times 6}{4}$$

$$\Rightarrow V_2 = 15 \text{ V}$$

Q4. A capacitor in steady state receives a charging current of 2A for 3ms out of a 10ms period. Find the discharging current magnitude for the remaining 7ms, assuming it is constant.

From Current second balance :-

$$\int_0^T i_C(t) dt = 0$$

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$$\Rightarrow 2(3 \times 10^{-3}) + (-I_2)(7 \times 10^{-3}) = 0$$

$$\Rightarrow I_2 = \frac{2 \times 3}{7}$$

$$\Rightarrow I_2 = 0.857 \text{ A}$$

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